Low-field carrier mobility models for 2D electron gas (2DEG)

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This documentation is part of `mobilitypy` package.[1] The following models for low-field electron mobility (μ) calculations from different scattering mechanism are based on the Reference-[2, 3, 4, 5, 6, 7, 8, 9, 10] and the references therein.

The available mobility models available are:

- 'v1' (available for v1.0.0 onwards). The 'Bassaler' model in v1.0.0 has been renamed to 'v1' as of v1.0.5 for consistence.
- 'v2' (only available for v1.0.5 onwards)
  - Several modifications has been incorporated in 'v2', including
    1. Function arguments units are made more user/field friendly (see the documentation).
    2. Added 'pseudomorphic strain' functionality. pseudomorphic strain modifies a_0 and c_0 according to the biaxial strain before passing these values to scattering mechanisms.
    3. Added Fang-Howard (FW) form-factor in dislocation scattering mechanisms. The FW form factors for acoustic phonons are corrected.
    4. Added scattering mechanism due to the strain field from the threading edge dislocation.
    5. Formula for WZ Ω_0 $\left[= \frac{\sqrt{3}}{2} a_0^2 c_0\right]$ in alloy disorder scattering equation is corrected to $\left[= \frac{\sqrt{3}}{8} a_0^2 c_0\right]$. Refer to Ref.[11].
       Ω_0 in AD scattering does not refer 'literally' to the unit cell volume, rather it refers to the effective volume that atom A (or B) occupies in the crystal. Only when each unit cell contains one atom, then this is equivalent to the unit cell volume. Historically the original derivation of alloy disorder scattering

rate assumed one atom per unit cell. Ω_0 was thus associated with unit cell volume. See, the derivations in Refs.[12] and [11] .
NOTE: The misinterpretation of the term showed up in Refs.[8, 9, 10] who used the wrong $\Omega_0 = \frac{\sqrt{3}}{2}a_0^2c_0$ value in the AD scattering. This is primitive unit cell volume of WZ crystal. But WZ primitive cell has 4 atoms per primitive unit cell. Therefore, correct Ω_0 should be $= \frac{\sqrt{3}}{8}a_0^2c_0$ for WZ structure.

Total mobility:

$$\mu = \left(\frac{m^*}{e}\sum_i \frac{1}{\tau_i}\right)^{-1}$$

where τ_i's are the momentum relaxation rate from different scattering mechanisms presented below.

1 Parameters

Parameters name	Symbol	SI Unit
Electron elementary charge	e	C
Boltzmann constant	k_B	$\mathrm{J\,K^{-1}}$
Temperature	T	K
Mass density	ρ	$\mathrm{kg\,m^{-3}}$
Lattice parameter	a_0	Å
Lattice parameter	c_0	Å
Unit cell volume	Ω_0	$\mathrm{Å}^3$
Bandgap energy	E_g	eV
Alloy-disordered scattering potential	U_0	eV
Electron effective mass	m^*	m_0
Electron rest mass	m_0	kg
Static dielectric constant	ϵ_s	ϵ_0
High frequency dielectric constant	ϵ_h	ϵ_0
Vacuum permittivity	ϵ_0	$\mathrm{C\,V^{-1}\,m^{-1}}$
Longitudinal acoustic phonon velocity	v_{LA}	$\mathrm{m\,s^{-1}}$
Transversal acoustic phonon velocity	v_{TA}	$\mathrm{m\,s^{-1}}$
Conduction band deformation potential	E_D	eV
Polar optical phonon energy	E_{POP}	eV
Electromechanical coupling coefficient	K^2	-
Elastic constants	C_{ij}	GPa
Scattering wave vector	k	$\mathrm{cm^{-1}}$
Fermi wave vector	k_F	$10^6\ \mathrm{cm^{-1}}$
Thomas-Fermi wave vector	q_{TF}	$10^6\ \mathrm{cm^{-1}}$
Fang-Howard variational WF parameter	b	$10^6\ \mathrm{cm^{-1}}$
Fang-Howard WF form factor	G, F	-
POP wave vector	k_0	$10^6\ \mathrm{cm^{-1}}$
2D electron gas density	n_{2D}	$10^{12}\ \mathrm{cm^{-2}}$
Interface RMS roughness	Δ	nm
Interface roughness correlation length	L	nm
Dislocation density	n_{DIS}	$10^8\ \mathrm{cm^{-2}}$
Dislocation occupancy	f_{DIS}	-
Mobility	μ	$\mathrm{cm^{-2}\,V^{-1}\,s^{-1}}$

Note: Latest units in the table follow ≥v2 model convention.

2 Basic theory of mobility

NOTE: This part of the discussion is adapted from Ref.[5], sec-6.2.5 & Ref.[13], Chapter-8.

Neglecting localization effect (breaks down for heavily disordered systems) and hence the electron can be assumed to adapt plane-wave wavefunction form for band-transport in 2DEG. Then the WF is of the form:

$$\langle \mathbf{r}|\mathbf{k}\rangle = \frac{1}{\sqrt{A}}\, e^{i\mathbf{k}\cdot\mathbf{r}}\, \chi(z)\, u_{n\mathbf{k}}(\mathbf{r}) \tag{1}$$

where $\mathbf{r}$ is the 2D in-plane vector and z is the perpendicular to the plane. Using Fermi's Golden rule, the scattering rate between state $\mathbf{k}$ and $\mathbf{k}'$:

$$\begin{aligned} \frac{1}{\tau(\mathbf{k}\to\mathbf{k}')} &= \frac{2\pi}{\hbar}|H_{\mathbf{k},\mathbf{k}'}|^2\, \delta(E_{\mathbf{k}} - E_{\mathbf{k}'}) \\ |H_{\mathbf{k},\mathbf{k}'}| &= \langle \mathbf{k}'|V(\mathbf{r},z)|\mathbf{k}\rangle \cdot I_{\mathbf{k},\mathbf{k}'} \end{aligned} \tag{2}$$

where $V(\mathbf{r},z)$ is the scattering potential, $I_{\mathbf{k},\mathbf{k}'}$ is the matrix element between lattice-periodic Bloch functions and $I_{\mathbf{k},\mathbf{k}'} \approx 1$ for wide bandgap materials.

From Boltzmann transport equation (BTE), the momentum relaxation rate of a single particle state $|\mathbf{k}\rangle$ is given by:

$$\frac{1}{\tau_m(\mathbf{k})} = N_{2d}^{sct} \sum_{\mathbf{k}'} \frac{1}{\tau(\mathbf{k}\to\mathbf{k}')}(1-\cos\theta) \tag{3}$$

where N_{2d}^{sct} is the total number of scatterers (e.g., defect centers, dislocation centers, impurity centers etc.) in the 2D area A and θ is the angle of scattering.

Assuming dilute concentration of scatterers (neglecting interactions between scattering centers, independent scatterers), and assuming all transport happens in the Fermi circle (degenerate metallic nature of the carrier; subbands with $E < E_F$ are filled and do not contribute to transport):

$$|\mathbf{k}| = |\mathbf{k}'| \approx k_{_F} \tag{4}$$

$$q = |\mathbf{q}| = |\mathbf{k}-\mathbf{k}'| = 2\, k_{_F}\, \sin(\theta/2) \tag{5}$$

$$\implies 1-\cos\theta = \frac{q^2}{2\, k_{_F}^2} \tag{6}$$

Therefore, for degenerate carriers, converting summation to integration over quasi-continuous $|\mathbf{k}\rangle$ states [see Ref.[13] for derivation]:

$$
\begin{aligned}
|H_{\mathbf{k},\mathbf{k}'}| &= \langle \mathbf{k}'|V(\mathbf{r},z)|\mathbf{k}\rangle \\
&= \frac{1}{A}\,\tilde{V}(\mathbf{q}) && (7)\\
\frac{1}{\tau(\mathbf{k}\to\mathbf{k}')} &= \frac{1}{A^2}\,\frac{2\pi}{\hbar}|\tilde{V}(\mathbf{q})|^2\,\delta(E_{\mathbf{k}}-E_{\mathbf{k}'}) && (8)\\
\frac{1}{\tau_m(\mathbf{k})} &= N_{2d}^{sct}\sum_{\mathbf{q}}\frac{1}{A^2}\,\frac{2\pi}{\hbar}|\tilde{V}(\mathbf{q})|^2\,\delta(E_{\mathbf{k}}-E_{\mathbf{k}'})\,(1-\cos\theta) \\
\sum_{\mathbf{q}} &\to \frac{A}{(2\pi)^2}\int d^2\mathbf{q} && (9)\\
\frac{1}{\langle\tau_m\rangle} &= N_{2d}^{sct}\frac{A}{(2\pi)^2}\int\frac{1}{A^2}\,\frac{2\pi}{\hbar}|\tilde{V}(\mathbf{q})|^2\,\delta(E_{\mathbf{k}}-E_{\mathbf{k}'})\,(1-\cos\theta)\,d^2\mathbf{q} \\
&= (A\,n_{2d}^{sct})\frac{A}{(2\pi)^2}\int\frac{1}{A^2}\,\frac{2\pi}{\hbar}|\tilde{V}(\mathbf{q})|^2\,\delta(E_{\mathbf{k}}-E_{\mathbf{k}'})\,\frac{\mathbf{q}^2}{2\,k_F^2}\,d^2\mathbf{q} \\
&= \frac{n_{2d}^{sct}}{2\pi\hbar}\int\frac{|\mathbf{q}|^2}{2\,k_F^2}|\tilde{V}(\mathbf{q})|^2\,\delta(E_{\mathbf{k}}-E_{\mathbf{k}'})\,d^2\mathbf{q} \\
&= n_{2d}^{sct}\,\frac{m^*}{2\pi\hbar^3k_F^3}\int_0^{2k_F}|\tilde{V}(q)|^2\,\frac{q^2}{\sqrt{1-(\frac{q}{2k_F})^2}}\,dq && (10)
\end{aligned}
$$

where $\tilde{V}(\mathbf{q})$ is the 2d Fourier transform scattering potential, n_{2d}^{sct} is density of scatterers per unit area, $k_F = \sqrt{2\pi\,n_s}$ is the Fermi wavevector, n_s is the 2DEG density.

3 Scattering mechanisms

$$
\begin{aligned}
b &= \left[\frac{33m^*e^2n_{2D}}{8\epsilon_s\hbar^2}\right]^{\frac{1}{3}} && (11)\\
k &= 2k_F u && (12)\\
k_F &= (2\pi n_{2D})^{1/2} && (13)\\
q_{TF} &= \frac{m^*e^2}{2\pi\epsilon_s\hbar^2} && (14)
\end{aligned}
$$

Unit analysis (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$
\begin{aligned}
n_{_{2D}} &= n'_{_{2D}} \times 10^{12}\ \mathrm{cm}^{-2} && (15)\\
k_{_F} &= (2\pi n_{_{2D}})^{1/2} = 2.51 \times (n'_{_{2D}})^{1/2}\ 10^6\,\mathrm{cm}^{-1} && (16)\\
q_{_{TF}} &= \frac{e^2\,m^*}{2\pi\epsilon_s\hbar^2} = \frac{e^2\,m_0}{2\pi\epsilon_0\hbar^2} \times 10^{-2} \times \frac{m^{*'}}{\epsilon'_s}\,\mathrm{cm}^{-1}\\
&= 377.945 \times \frac{m^{*'}}{\epsilon'_s}\,10^6\,\mathrm{cm}^{-1} && (17)\\
\frac{q_{_{TF}}}{2\,k_{_F}} &= \frac{e^2\,m^*}{2\pi\epsilon_s\hbar^2} \times \frac{1}{2} \times \frac{1}{(2\pi n_{_{2D}})^{1/2}} = \frac{e^2\,m^*}{2(2\pi)^{\frac{3}{2}}\epsilon_s\hbar^2\,n_{2d}^{\frac{1}{2}}}\\
&= \frac{e^2\,m_0}{2(2\pi)^{\frac{3}{2}}\epsilon_0\hbar^2 \times 10^6 \times 10^2} \times \frac{m^{*'}}{\epsilon'_s\,n_{2d}^{'\frac{1}{2}}}\\
&= 75.389 \times \frac{m^{*'}}{\epsilon'_s\,n_{2d}^{'\frac{1}{2}}} && (18)\\
b &= \left(\frac{33m^*e^2n_{_{2D}}}{8\epsilon_s\hbar^2}\right)^{\frac{1}{3}}\\
&= \left(\frac{33m_0e^2}{8\epsilon_0\hbar^2} \times 10^{12}\,cm^{-2}\right)^{\frac{1}{3}} \times \left(\frac{m^{*'}\,n'_{_{2D}}}{\epsilon'_s}\right)^{\frac{1}{3}}\\
&= 21.397 \times \left(n'_{_{2D}} \times \frac{m^{*'}}{\epsilon'_s}\right)^{\frac{1}{3}}\ 10^6\,\mathrm{cm}^{-1} && (19)
\end{aligned}
$$

3.1 Structural related scattering

3.1.1 Interface roughness momentum relaxation rate

Implementation is based on Refs.[5, 9, 10].
NOTE: The pre-factors for this scattering mechanism are different in Ref.[3, 6] which are again different from Ref.[11]. Here, we use Ref.[5].

The momentum relaxation rate:

$$\frac{1}{\tau_{IFR}} = \frac{e^4 m^*(\Delta L n_{2D})^2}{8\,\epsilon_s^2\hbar^3} \times \int_0^1 \frac{u^4\, e^{-(Lk_F u)^2}}{\left(u + \frac{q_{TF}}{2k_F}G(u)\right)^2 \sqrt{1-u^2}} du \tag{20}$$

$$G(u) = \frac{2\eta(u)^3 + 3\eta(u)^2 + 3\eta(u)}{8} \tag{21}$$

$$\eta(u) = \frac{b}{b + 2k_F u} \tag{22}$$

Unit analysis (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned}
\frac{1}{\tau_{IFR}} &= \frac{e^4 m^*(\Delta L n_{2D})^2}{8\,\epsilon_s^2\hbar^3} \times \int_0^1 \frac{u^4\, e^{-(Lk_F u)^2}}{\left(u + \frac{q_{TF}}{2k_F}G(u)\right)^2 \sqrt{1-u^2}} du \\
&= \frac{e^4 m_0}{8\epsilon_0^2\hbar^3} \times 10^{-4} \times \frac{m^{*'}(\Delta' L' n'_{2D})^2}{\epsilon_s'^2} \times \int_0^1 \frac{u^4\, e^{-(Lk_F u)^2}}{\left(u + \frac{q_{TF}}{2k_F}G(u)\right)^2 \sqrt{1-u^2}} du \\
&= 81.605 \times \frac{m^{*'}}{\epsilon_s'^2} \times (\Delta' L' n'_{2D})^2 \times \int_0^1 \frac{u^4\, e^{-(L' k'_F \times 10^{-1} \times u)^2}}{\left(u + \frac{q_{TF}}{2k_F}G(u)\right)^2 \sqrt{1-u^2}} du \\
&\qquad 10^{12}\ \mathrm{s}^{-1}
\end{aligned} \tag{23}$$

3.1.2 Dislocation momentum relaxation rate

3.1.2.1 Scattering from line charge of threading dislocation core

Implementation is based on Refs.[11, 13, 5, 14, 9, 10].
NOTE: Refs.[11, 13] includes the Fang-Howard form factors in the expression but Refs.[5, 14, 9, 10] do not. The momentum relaxation rate:

$$\frac{1}{\tau_{DIS}} = \frac{e^4\, m^* n_{DIS} f_{DIS}^2}{4\pi\epsilon_s^2\hbar^3 k_F^4 c_0^2} \times \int_0^1 \frac{F(u)^2}{\left(u + \frac{q_{TF}}{2k_F}G(u)\right)^2 \sqrt{1-u^2}} du \tag{24}$$

$$F(u) = \eta(u)^3 \tag{25}$$

$$G(u) = \frac{2\eta(u)^3 + 3\eta(u)^2 + 3\eta(u)}{8} \tag{26}$$

$$\eta(u) = \frac{b}{b + 2k_F u} \tag{27}$$

Unit analysis (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned}
\frac{1}{\tau_{_{DIS}}} &= \frac{e^4\, m^* n_{_{DIS}} f_{_{DIS}}^2}{4\pi\epsilon_s^2\hbar^3 k_{_F}^4 c_0^2} \times \int_0^1 \frac{F(u)^2}{\left(u+\frac{q_{_{TF}}}{2k_{_F}}G(u)\right)^2\sqrt{1-u^2}}du \\
&= \frac{e^4\, m_0}{4\pi\epsilon_0^2\hbar^3}\frac{10^8\, cm^{-2}}{(10^6 cm^{-1})^4\,(10^{-8}cm)^2}\frac{m^{*'}}{\epsilon_s'^2}\frac{n'_{_{DIS}} f_{_{DIS}}^2}{c_0'^2 k_{_F}'^4} \\
&\qquad \times \int_0^1 \frac{F(u)^2}{\left(u+\frac{q_{_{TF}}}{2k_{_F}}G(u)\right)^2\sqrt{1-u^2}}du \\
&= 519511.019 \times \frac{m^{*'}}{\epsilon_s'^2}\frac{n'_{_{DIS}} f_{_{DIS}}^2}{c_0'^2 k_{_F}'^4} \times \int_0^1 \frac{F(u)^2}{\left(u+\frac{q_{_{TF}}}{2k_{_F}}G(u)\right)^2\sqrt{1-u^2}}du \\
&\qquad 10^{12}\ \mathrm{s}^{-1} \qquad (28)
\end{aligned}$$

3.1.2.2 Scattering from strain field of threading dislocation

This form of scattering arises due to the presence of dislocations, regardless of the presence or absence of charges at the core.

2.1.2.2.A Due to threading edge dislocation strain field

This scattering mechanism is implemented based on Refs. [11, 13, 7, 4, 5, 2]. **NOTE:** Refs.[11, 13] includes the Fang-Howard form factors in the expression but other Refs do not. The momentum relaxation rate:

$$\frac{1}{\tau_{_{DIS}}^{str}} = \frac{m^* n_{_{DIS}} b_e^2 E_D^2}{2\pi k_{_F}^2\hbar^3}\left(\frac{1-2\gamma}{1-\gamma}\right)^2 \times \int_0^1 \frac{u^2\, F(u)^2}{\left(u+\frac{q_{_{TF}}}{2k_{_F}}G(u)\right)^2\sqrt{1-u^2}}du \qquad (29)$$

$$F(u) = \eta(u)^3 \qquad (30)$$

$$G(u) = \frac{2\eta(u)^3+3\eta(u)^2+3\eta(u)}{8} \qquad (31)$$

$$\eta(u) = \frac{b}{b+2k_{_F}u} \qquad (32)$$

$$b_e = \frac{1}{3} < 11\overline{20} >= a_0 \qquad (33)$$

$$\gamma = \frac{C_{12}}{C_{11}+C_{12}} \quad [\text{"Isotropic Poisson's ratio"}] \qquad (34)$$

NOTE-1: For WZ structure the Poisson ratio is anisotropic[15]. However, the strain field due to edge dislocation uses 'isotropic/average Poisson ratio'. We use the definition $\gamma = C_{12}/(C_{11}+C_{12})$ following reference Shi et al., APL 74, 573 (1999)[2]. This was also used in the thesis.[5]
NOTE-2: In Ref.[15] there is a printing mistake. The s_{11} expression should be $2s_{11} = sc_{33} + \frac{1}{c_{11}-c_{12}}$.
Unit analysis (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned}
\frac{1}{\tau^{str}_{DIS}} &= \frac{m^* n_{DIS} b_e^2 E_D^2}{2\pi k_F^2 \hbar^3}\left(\frac{1-2\gamma}{1-\gamma}\right)^2 \int_0^1 \frac{u^2\, F(u)^2}{\left(u+\frac{q_{TF}}{2k_F}G(u)\right)^2\sqrt{1-u^2}}du \\
&= \frac{m_0}{2\pi\hbar^3}\frac{10^8\, cm^{-2}\,(10^{-8}\, cm)^2\, e'^2\, J^2}{(10^6\, cm^{-1})^2}\frac{m^{*'} n'_{DIS} a_0'^2 E_D'^2}{k_F'^2}\left(\frac{1-2\gamma}{1-\gamma}\right)^2 \\
&\qquad \times \int_0^1 \frac{u^2\, F(u)^2}{\left(u+\frac{q_{TF}}{2k_F}G(u)\right)^2\sqrt{1-u^2}}du \\
&= 0.00317 \times \frac{m^{*'} n'_{DIS} a_0'^2 E_D'^2}{k_F'^2}\left(\frac{1-2\gamma}{1-\gamma}\right)^2 \\
&\qquad \times \int_0^1 \frac{u^2\, F(u)^2}{\left(u+\frac{q_{TF}}{2k_F}G(u)\right)^2\sqrt{1-u^2}}du \quad 10^{12}\ \mathrm{s}^{-1}
\end{aligned} \tag{35}$$

2.1.2.2.B Due to threading screw dislocation strain field
Screw dislocation strain scattering has not yet been implemented.

3.1.3 Alloy-disordered momentum relaxation rate

This scattering mechanism is implemented based on Refs. [11, 5, 2, 9, 10]. The momentum relaxation rate:

$$\begin{aligned}
\frac{1}{\tau_{_{AD}}} &= \frac{m^*\Omega_0 U_0^2 x(1-x)}{\hbar^3} \times \int_{-\infty}^{\infty} \Psi(z)^4\, dz \\
&= \frac{m^*\Omega_0 U_0^2 x(1-x)}{\hbar^3} \times \int_{-\infty}^{0} \Psi(z)^4\, dz \times \int_{0}^{\infty} \Psi(z)^4\, dz \\
&\approx \frac{m^*\Omega_0 U_0^2 x(1-x)}{\hbar^3} \times \int_{0}^{\infty} \Psi(z)^4\, dz \text{ [assuming channel 2DEG only]} \\
&= \frac{m^*\Omega_0 U_0^2 x(1-x)}{\hbar^3} \frac{3b}{16} \qquad (36)
\end{aligned}$$

NOTE-1: Here, the total wavefunction is separated into barrier and channel components at the barrier–channel interface ($z = 0$). We then neglect any alloy-scattering contributions originating from the barrier. This approximation is justified because the wavefunction penetration into the barrier is negligible, as demonstrated in Refs. [9, 10, 16] Using Fang-Howard approximation the integral of the wavefunction in the channel become $\int_0^\infty \Psi(z)^4\, dz = \frac{3\,b}{16}$.
NOTE-2: Barrier contribution can also be implemented following Refs.[5, 2, 16]. The implementation is not available yet. Contact developers.
Unit analysis (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned}
\frac{1}{\tau_{_{AD}}} &= \frac{m^*\Omega_0 U_0^2 x(1-x)}{\hbar^3} \frac{3b}{16} \\
&= \frac{3\, m_0}{16\, \hbar^3} (10^{-8}\, cm^{-1})^3\, (e'^2\, J^2)\, (10^6\, cm^{-1}) \times [m^{*'} \Omega_0' U_0'^2 x(1-x)\, b'] \\
&= 0.37384 \times m^{*'} \Omega_0' U_0'^2 x(1-x)\, b' \quad 10^{12}\ \mathrm{s}^{-1} \qquad (37)
\end{aligned}$$

3.2 Phonon related scattering

3.2.1 Deformation potential scattering momentum relaxation rate

This scattering mechanism is implemented based on Refs. [5, 6, 3, 9, 10].
NOTE: The FW form factors were wrong in Refs.[9, 10]. The expression

in those refs. were adapted from Refs.[6, 3]. Refs.[6, 3] wrongly mixed the F_{11} notation with $G(u)$. Tracing back to their original derivations for correct notations we find that F_{11} notation in Refs.[6, 3] should actually be interpreted as $G(u)$ in our case. This correction has been done in 'v2' model on wards. We emphasize that since the Form factors were very close to 1 already, the effect were negligible on the results presented in Refs.[9, 10].
The momentum relaxation rate:

$$\begin{aligned}
\frac{1}{\tau_{DP}} &= \frac{3m^* E_D^2 k_B T\, b}{16\,\rho v_{LA}^2 \hbar^3}\frac{4}{\pi} \times \int_0^1 \frac{u^4}{\left(u + \frac{q_{TF}}{2k_F}G(u)\right)^2 \sqrt{1-u^2}} du \\
&= \frac{3m^* E_D^2 k_B T\, b}{4\pi\rho v_{LA}^2 \hbar^3} \times \int_0^1 \frac{u^4}{\left(u + \frac{q_{TF}}{2k_F}G(u)\right)^2 \sqrt{1-u^2}} du \qquad (38)\\
G(u) &= \frac{2\eta(u)^3 + 3\eta(u)^2 + 3\eta(u)}{8} \qquad (39)\\
\eta(u) &= \frac{b}{b + 2k_F u} \qquad (40)
\end{aligned}$$

Unit analysis (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned}
\frac{1}{\tau_{DP}} &= \frac{3m^* E_D^2 k_B T\, b}{4\pi\rho v_{LA}^2 \hbar^3} \times \int_0^1 \frac{u^4}{\left(u + \frac{q_{TF}}{2k_F}G(u)\right)^2 \sqrt{1-u^2}} du \\
&= \frac{3m_0 k_B}{4\pi\hbar^3} e'^2\, 10^6 \frac{E_D'^2 T' b'}{\rho' v_{LA}'^2} \times \int_0^1 \frac{u^4}{\left(u + \frac{q_{TF}}{2k_F}G(u)\right)^2 \sqrt{1-u^2}} du \\
&= 6571673.424 \times \frac{E_D'^2 T' b'}{\rho' v_{LA}'^2} \times \int_0^1 \frac{u^4}{\left(u + \frac{q_{TF}}{2k_F}G(u)\right)^2 \sqrt{1-u^2}} du \\
&\qquad 10^{12}\ \mathrm{s}^{-1} \qquad (41)
\end{aligned}$$

3.2.2 Piezoelectric phonon scattering momentum relaxation rate

This scattering mechanism is implemented based on Refs. [6, 3, 9, 10].
NOTE: The FW form factors were wrong in Refs.[9, 10]. The expression in those refs. were adapted from Refs.[6, 3]. Refs.[6, 3] wrongly mixed the F_{11} notation with $G(u)$. Tracing back to their original derivations for

correct notations we find that F_{11} notation in Refs.[6, 3] should actually be interpreted as $G(u)$ in our case. This correction has been done in 'v2' model on wards. We emphasize that since the Form factors were very close to 1 already, the effect were negligible on the results presented in Refs.[9, 10].
The momentum relaxation rate:

$$\frac{1}{\tau_{PE}} = \frac{m^* e^2 K^2 k_B T}{\pi \epsilon_s \hbar^3 k_F} \times \int_0^1 \frac{u^3 G(u)}{\left(u + \frac{q_{TF}}{2k_F} G(u)\right)^2 \sqrt{1-u^2}} du \tag{42}$$

$$G(u) = \frac{2\eta(u)^3 + 3\eta(u)^2 + 3\eta(u)}{8} \tag{43}$$

$$\eta(u) = \frac{b}{b + 2k_F u} \tag{44}$$

Unit analysis (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned}
\frac{1}{\tau_{PE}} &= \frac{m^* e^2 K^2 k_B T}{\pi \epsilon_s \hbar^3 k_F} \times \int_0^1 \frac{u^3\, G(u)}{\left(u + \frac{q_{TF}}{2k_F} G(u)\right)^2 \sqrt{1-u^2}} du \\
&= \frac{m_0 e^2 k_B}{\pi \epsilon_0 \hbar^3} \frac{1}{10^6} \frac{m^{*\prime} K^2 T'}{\epsilon_s' k_F'} \times \int_0^1 \frac{u^3\, G(u)}{\left(u + \frac{q_{TF}}{2k_F} G(u)\right)^2 \sqrt{1-u^2}} du \\
&= 98.961 \times \frac{m^{*\prime} K^2 T'}{\epsilon_s' k_F'} \times \int_0^1 \frac{u^3\, G(u)}{\left(u + \frac{q_{TF}}{2k_F} G(u)\right)^2 \sqrt{1-u^2}} du \\
&\qquad 10^{12}\ \mathrm{s}^{-1}
\end{aligned} \tag{45}$$

3.2.3 Acoustic phonon scattering relaxation rate

$$\frac{1}{\tau_{AP}} = \frac{1}{\tau_{DP}} + \frac{1}{\tau_{PE}} \tag{46}$$

3.2.4 Polar optical phonon scattering momentum relaxation rate

This scattering mechanism is implemented based on Refs. [17, 11, 9, 10].
NOTE: The expression for b is different in Ref.[17] from that is used here.

The momentum relaxation rate:

$$
\begin{aligned}
\frac{1}{\tau_{_{POP}}} &= \frac{e^2 m^* E_{_{POP}} G(k_0)}{2\epsilon^* k_0 \hbar^3} \times \frac{y\,(1+y-e^{-y})^{-1}}{e^{\frac{E_{POP}}{k_B T}}-1} \\
&= \frac{e^2 m^* E_{_{POP}} G(k_0)\hbar}{2\epsilon^* \sqrt{2}\sqrt{m^* E_{_{POP}}}\hbar^3} \times \frac{y\,(1+y-e^{-y})^{-1}}{e^{\frac{E_{POP}}{k_B T}}-1} \\
&= \frac{e^2 \sqrt{m^* E_{_{POP}}} G(k_0)}{2\sqrt{2}\,\epsilon^* \hbar^2} \times \frac{y\,(1+y-e^{-y})^{-1}}{e^{\frac{E_{POP}}{k_B T}}-1} \qquad (47) \\
G(k_0) &= \frac{2\eta(k_0)^3 + 3\eta(k_0)^2 + 3\eta(k_0)}{8} \qquad (48) \\
\eta(k_0) &= \frac{b}{b+k_0} \qquad (49) \\
k_0 &= \sqrt{\frac{2m^* E_{_{POP}}}{\hbar^2}} \qquad (50) \\
\frac{1}{\epsilon^*} &= \frac{1}{\epsilon_h} - \frac{1}{\epsilon_s} \qquad (51) \\
y &= \frac{\pi \hbar^2 n_{_{2D}}}{m^* k_{_B} T} \qquad (52)
\end{aligned}
$$

Unit analysis (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$
\begin{aligned}
k_0 &= \sqrt{\frac{2m^* E_{_{POP}}}{\hbar^2}} = \sqrt{\frac{2m_0 e'}{\hbar^2}}\sqrt{m^{*\prime} E'_{_{POP}}} \\
&= 51.232 \times \sqrt{m^{*\prime} E'_{_{POP}}}\; 10^6 \;\; \mathrm{cm}^{-1} \qquad (53) \\
y &= \frac{\pi \hbar^2 n_{_{2D}}}{m^* k_{_B} T} = \frac{\pi \hbar^2 \times 10^{12}}{m_0\, k_{_B}} \frac{n'_{_{2D}}}{m^{*\prime} T'} \\
&= 27.7798 \times \frac{n'_{_{2D}}}{m^{*\prime} T'} \qquad (54)
\end{aligned}
$$

$$\begin{aligned}
\frac{1}{\tau_{POP}} &= \frac{e^2\sqrt{m^* E_{POP}}G(k_0)}{2\sqrt{2}\,\epsilon^*\hbar^2} \times \frac{y\,(1+y-e^{-y})^{-1}}{e^{\frac{E_{POP}}{k_B T}}-1} \\
&= \frac{e^2\sqrt{m_0\,e'}}{2\sqrt{2}\,\epsilon_0\hbar^2}\frac{\sqrt{m^{*\prime}E'_{POP}}G(k_0)}{\epsilon^*} \times \frac{y\,(1+y-e^{-y})^{-1}}{e^{\frac{e'}{k_B}\frac{E'_{POP}}{T'}}-1} \\
&= 35210.682 \times \frac{\sqrt{m^{*\prime}E'_{POP}}G(k_0)}{\epsilon^*} \times \frac{y\,(1+y-e^{-y})^{-1}}{e^{11604.52\times\frac{E'_{POP}}{T'}}-1} \\
&\qquad 10^{12}\ \mathrm{s}^{-1} \qquad (55)
\end{aligned}$$

4 Other properties

4.1 Total mobility

Total mobility:

$$\mu = \left(\frac{m^*}{e}\sum_i \frac{1}{\tau_i}\right)^{-1} \qquad (56)$$

where τ_i's are the momentum relaxation rate from different scattering mechanisms presented below. **Unit analysis** (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned}
\mu &= \left(\frac{m^*}{e}\sum_i \frac{1}{\tau_i}\right)^{-1} \\
&= \left(\frac{m_0}{e} \times m^{*\prime} \times 10^{12}\,s^{-1} \times \sum_i \frac{1}{\tau'_i}\right)^{-1} \\
&= \frac{10^4\,m^{-2}.cm^2}{\left(5.686\times 10^{-12}\,V.m^{-2}.s^2 \times m^{*\prime} \times 10^{12}\,s^{-1} \times \sum_i \frac{1}{\tau'_i}\right)} \\
&= \frac{10^4}{\left(5.686 \times m^{*\prime} \times \sum_i \frac{1}{\tau'_i}\right)}\ \mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1} \qquad (57)
\end{aligned}$$

4.2 Figure-of-merit

4.2.1 Lateral figure-of-merit (LFOM)

Implemented based on Ref.[10].

$$\begin{aligned} \text{LFOM} &= e \times n_{2d} \times \mu \times E_{cr}^2 && (58) \\ E_{cr} &= 1.73 \times 10^5 \times E_g^{2.5}\ \mathrm{V\,cm^{-1}}\ \text{[for direct bandgap]} && (59) \\ &= 2.38 \times 10^5 \times E_g^2\ \mathrm{V\,cm^{-1}}\ \text{[for direct bandgap]} && (60) \end{aligned}$$

Relation between the critical electric field (E_{cr} V/cm) and bandgap value (E_g eV) is taken from Ref.[18].

Unit analysis (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned} \text{LFOM} &= e \times n_{2d} \times \mu \times E_{cr}^2 \\ &= e \times n'_{2d} \times 10^{12}\, cm^{-2} \times \mu' \times cm^2.V^{-1}.s^{-1} \\ &\qquad \times 1.73^2 \times 10^{10} \times E_g^5\, (V.cm^{-2})^2 \\ &= 4795.15 \times n'_{2d}\, \mu'\ E_g^5\ \ C.V.s^{-1}.cm^{-2} \\ &= 4795.15 \times n'_{2d}\, \mu'\ E_g^5\ W.cm^{-2} \quad [1\ \mathrm{W} = 1\ \mathrm{C.V/s}] \\ &= 0.00479515 \times n'_{2d}\, \mu'\ E_g^5\ MW.cm^{-2} && (61) \end{aligned}$$

4.3 Sheet resistance

Implemented based on Ref.[10]. Sheet resistance:

$$R_{sh} = \frac{1}{e\ n_{2d}\ \mu} \qquad (62)$$

Unit analysis (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned} R_{sh} &= \frac{1}{e\ n_{2d}\ \mu} \\ &= \frac{1}{e \times 10^{12}} \times \frac{1}{n'_{2d}\ \mu'} \\ &= 6241509.075 \times \frac{1}{n'_{2d}\ \mu}\ \Omega/\square && (63) \\ &\quad [1\ C/V = 1\ s/\Omega \implies 1\ \Omega = 1\ C^{-1}.V.s] \end{aligned}$$

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